



COMP30026

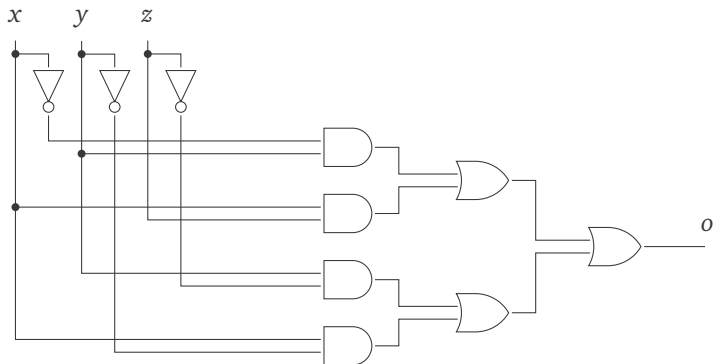
Models of Computation

Cezary Kaliszyk and William Umboh

Lecture 5

Recap and Predicate Logic

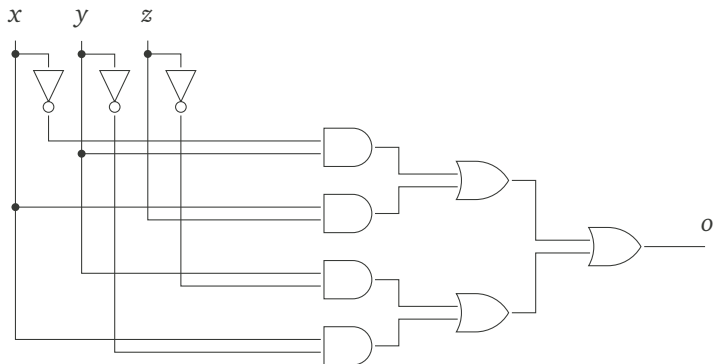
Recap: Encoding to Propositional Logic



Encode o :

o

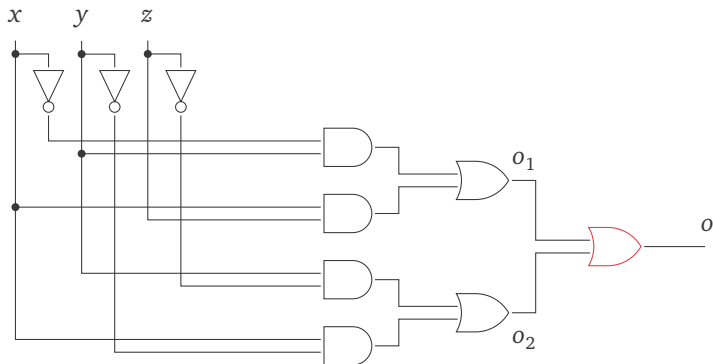
Recap: Encoding to Propositional Logic



Encode o :

o

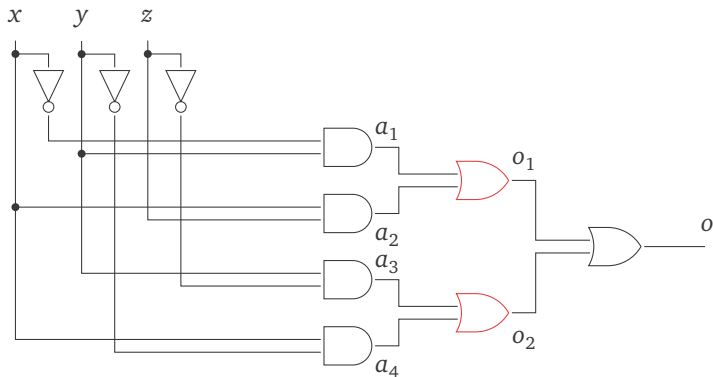
Recap: Encoding to Propositional Logic



Encode o :

$$o \wedge (o \leftrightarrow o_1 \vee o_2)$$

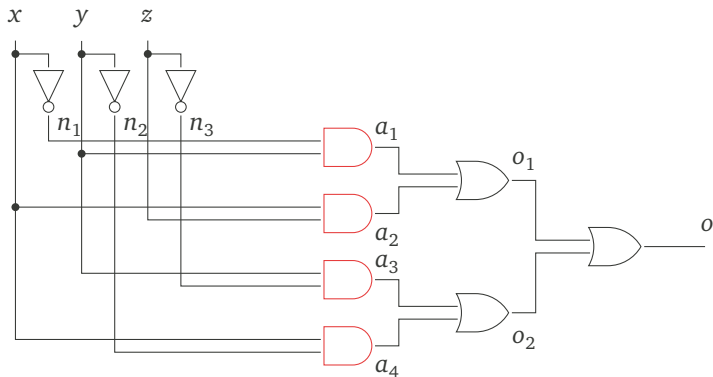
Recap: Encoding to Propositional Logic



Encode o :

$$o \wedge (o \leftrightarrow o_1 \vee o_2) \wedge (o_1 \leftrightarrow a_1 \vee a_2) \wedge (o_2 \leftrightarrow a_3 \vee a_4)$$

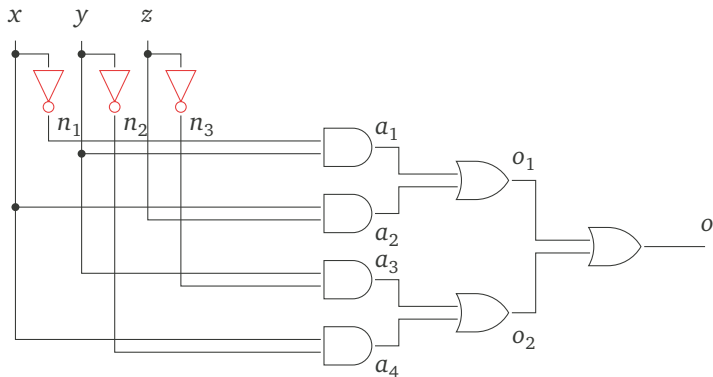
Recap: Encoding to Propositional Logic



Encode o :

$$o \wedge (o \leftrightarrow o_1 \vee o_2) \wedge (o_1 \leftrightarrow a_1 \vee a_2) \wedge (o_2 \leftrightarrow a_3 \vee a_4) \\ \wedge (a_1 \leftrightarrow n_1 \wedge y) \wedge (a_2 \leftrightarrow x \wedge z) \wedge (a_3 \leftrightarrow y \wedge n_3) \wedge (a_4 \leftrightarrow x \wedge n_2)$$

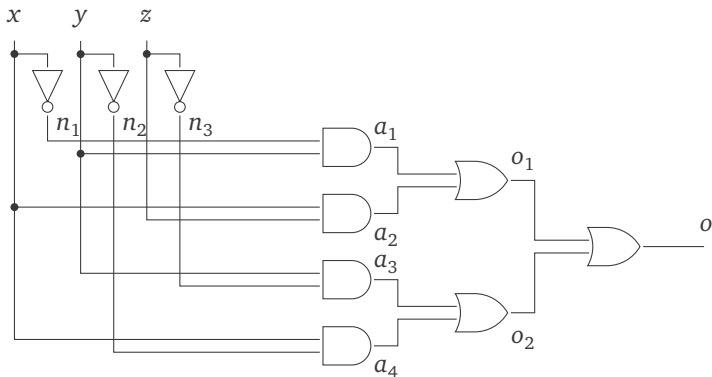
Recap: Encoding to Propositional Logic



Encode o :

$$\begin{aligned}
 & o \wedge (o \leftrightarrow o_1 \vee o_2) \wedge (o_1 \leftrightarrow a_1 \vee a_2) \wedge (o_2 \leftrightarrow a_3 \vee a_4) \\
 & \wedge (a_1 \leftrightarrow n_1 \wedge y) \wedge (a_2 \leftrightarrow x \wedge z) \wedge (a_3 \leftrightarrow y \wedge n_3) \wedge (a_4 \leftrightarrow x \wedge n_2) \\
 & \wedge (n_1 \leftrightarrow \neg x) \wedge (n_2 \leftrightarrow \neg y) \wedge (n_3 \leftrightarrow \neg z)
 \end{aligned}$$

Recap: Encoding to Propositional Logic



Encode o :

$$\begin{aligned}
 & o \wedge (o \leftrightarrow o_1 \vee o_2) \wedge (o_1 \leftrightarrow a_1 \vee a_2) \wedge (o_2 \leftrightarrow a_3 \vee a_4) \\
 & \wedge (a_1 \leftrightarrow n_1 \wedge y) \wedge (a_2 \leftrightarrow x \wedge z) \wedge (a_3 \leftrightarrow y \wedge n_3) \wedge (a_4 \leftrightarrow x \wedge n_2) \\
 & \wedge (n_1 \leftrightarrow \neg x) \wedge (n_2 \leftrightarrow \neg y) \wedge (n_3 \leftrightarrow \neg z)
 \end{aligned}$$



Recap: CNF

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge (Q \leftrightarrow S)$$



Recap: CNF

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge (Q \leftrightarrow S)$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((Q \rightarrow S) \wedge (S \rightarrow Q))$$



Recap: CNF

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge (Q \leftrightarrow S)$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((Q \rightarrow S) \wedge (S \rightarrow Q))$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$



Recap: CNF

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge (Q \leftrightarrow S)$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((Q \rightarrow S) \wedge (S \rightarrow Q))$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$(\neg(P \vee Q) \vee \neg(\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$



Recap: CNF

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge (Q \leftrightarrow S)$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((Q \rightarrow S) \wedge (S \rightarrow Q))$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$(\neg(P \vee Q) \vee \neg(\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$((\neg P \wedge \neg Q) \vee (\neg\neg P \wedge \neg R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$



Recap: CNF

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge (Q \leftrightarrow S)$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((Q \rightarrow S) \wedge (S \rightarrow Q))$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$(\neg(P \vee Q) \vee \neg(\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$((\neg P \wedge \neg Q) \vee (\neg\neg P \wedge \neg R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$((\neg P \wedge \neg Q) \vee (P \wedge \neg R)) \wedge (\neg Q \vee S) \wedge (\neg S \vee Q)$$



Recap: CNF

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge (Q \leftrightarrow S)$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((Q \rightarrow S) \wedge (S \rightarrow Q))$$

$$\neg((P \vee Q) \wedge (\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$(\neg(P \vee Q) \vee \neg(\neg P \vee R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$((\neg P \wedge \neg Q) \vee (\neg\neg P \wedge \neg R)) \wedge ((\neg Q \vee S) \wedge (\neg S \vee Q))$$

$$((\neg P \wedge \neg Q) \vee (P \wedge \neg R)) \wedge (\neg Q \vee S) \wedge (\neg S \vee Q)$$

$$(\neg P \vee P) \wedge (\neg P \vee \neg R) \wedge (\neg Q \vee P) \wedge (\neg Q \vee \neg R) \wedge (\neg Q \vee S) \wedge (\neg S \vee Q)$$



Recap: Resolution (1/3)

$$(\neg p \vee \neg q \vee r) \wedge (p \vee r) \wedge (q \vee r) \wedge \neg r$$

$$1 \quad \{\neg p, \neg q, r\}$$

$$2 \quad \{p, r\}$$

$$3 \quad \{q, r\}$$

$$4 \quad \{\neg r\}$$



Recap: Resolution (1/3)

$$(\neg p \vee \neg q \vee r) \wedge (p \vee r) \wedge (q \vee r) \wedge \neg r$$

$$1 \quad \{\neg p, \neg q, r\}$$

$$2 \quad \{p, r\}$$

$$3 \quad \{q, r\}$$

$$4 \quad \{\neg r\}$$



Recap: Resolution (1/3)

$$(\neg p \vee \neg q \vee r) \wedge (p \vee r) \wedge (q \vee r) \wedge \neg r$$

$$1 \quad \{\neg p, \neg q, r\}$$

$$2 \quad \{p, r\}$$

$$3 \quad \{q, r\}$$

$$4 \quad \{\neg r\}$$

$$5 \quad \{\neg q, r\} \quad \text{resolve 1, 2, } p$$



Recap: Resolution (1/3)

$$(\neg p \vee \neg q \vee r) \wedge (p \vee r) \wedge (q \vee r) \wedge \neg r$$

$$1 \quad \{\neg p, \neg q, r\}$$

$$2 \quad \{p, r\}$$

$$3 \quad \{q, r\}$$

$$4 \quad \{\neg r\}$$

$$5 \quad \{\neg q, r\} \quad \text{resolve 1, 2, } p$$

Recap: Resolution (1/3)

$$(\neg p \vee \neg q \vee r) \wedge (p \vee r) \wedge (q \vee r) \wedge \neg r$$

1 $\{\neg p, \neg q, r\}$

2 $\{p, r\}$

3 $\{q, r\}$

4 $\{\neg r\}$

5 $\{\neg q, r\}$ resolve 1, 2, p

6 $\{r\}$ resolve 3, 5, q



Recap: Resolution (1/3)

$$(\neg p \vee \neg q \vee r) \wedge (p \vee r) \wedge (q \vee r) \wedge \neg r$$

1 $\{\neg p, \neg q, r\}$

2 $\{p, r\}$

3 $\{q, r\}$

4 $\{\neg r\}$

5 $\{\neg q, r\}$ resolve 1, 2, p

6 $\{r\}$ resolve 3, 5, q



Recap: Resolution (1/3)

$$(\neg p \vee \neg q \vee r) \wedge (p \vee r) \wedge (q \vee r) \wedge \neg r$$

- 1 $\{\neg p, \neg q, r\}$
- 2 $\{p, r\}$
- 3 $\{q, r\}$
- 4 $\{\neg r\}$
- 5 $\{\neg q, r\}$ resolve 1, 2, p
- 6 $\{r\}$ resolve 3, 5, q
- 7 $\perp$ resolve 4, 6, r



Recap: Resolution (1/3)

$$(\neg p \vee \neg q \vee r) \wedge (p \vee r) \wedge (q \vee r) \wedge \neg r$$

- 1 $\{\neg p, \neg q, r\}$
- 2 $\{p, r\}$
- 3 $\{q, r\}$
- 4 $\{\neg r\}$
- 5 $\{\neg q, r\}$ resolve 1, 2, p
- 6 $\{r\}$ resolve 3, 5, q
- 7 $\perp$ resolve 4, 6, r

unsatisfiable



Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

- 1 $\{p\}$
- 2 $\{\neg p, q, r\}$
- 3 $\{\neg p, \neg q, \neg r\}$
- 4 $\{\neg p, s, t\}$
- 5 $\{\neg p, \neg s, \neg t\}$
- 6 $\{\neg s, q\}$
- 7 $\{r, t\}$
- 8 $\{\neg t, s\}$



Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

- 1 $\{p\}$
- 2 $\{\neg p, q, r\}$
- 3 $\{\neg p, \neg q, \neg r\}$
- 4 $\{\neg p, s, t\}$
- 5 $\{\neg p, \neg s, \neg t\}$
- 6 $\{\neg s, q\}$
- 7 $\{r, t\}$
- 8 $\{\neg t, s\}$

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

- 1 $\{p\}$
- 2 $\{\neg p, q, r\}$
- 3 $\{\neg p, \neg q, \neg r\}$
- 4 $\{\neg p, s, t\}$
- 5 $\{\neg p, \neg s, \neg t\}$
- 6 $\{\neg s, q\}$
- 7 $\{r, t\}$
- 8 $\{\neg t, s\}$
- 9 $\{\neg q, \neg r\}$ res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

- 1 $\{p\}$
- 2 $\{\neg p, q, r\}$
- 3 $\{\neg p, \neg q, \neg r\}$
- 4 $\{\neg p, s, t\}$
- 5 $\{\neg p, \neg s, \neg t\}$
- 6 $\{\neg s, q\}$
- 7 $\{r, t\}$
- 8 $\{\neg t, s\}$
- 9 $\{\neg q, \neg r\}$ res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$	10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$			
3	$\{\neg p, \neg q, \neg r\}$			
4	$\{\neg p, s, t\}$			
5	$\{\neg p, \neg s, \neg t\}$			
6	$\{\neg s, q\}$			
7	$\{r, t\}$			
8	$\{\neg t, s\}$			
9	$\{\neg q, \neg r\}$			res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$	10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$			
3	$\{\neg p, \neg q, \neg r\}$			
4	$\{\neg p, s, t\}$			
5	$\{\neg p, \neg s, \neg t\}$			
6	$\{\neg s, q\}$			
7	$\{r, t\}$			
8	$\{\neg t, s\}$			
9	$\{\neg q, \neg r\}$	res 1, 3, p		



Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$		
2	$\{\neg p, q, r\}$		
3	$\{\neg p, \neg q, \neg r\}$		
4	$\{\neg p, s, t\}$		
5	$\{\neg p, \neg s, \neg t\}$		
6	$\{\neg s, q\}$		
7	$\{r, t\}$		
8	$\{\neg t, s\}$		
9	$\{\neg q, \neg r\}$	res 1, 3,	p
10	$\{s, t\}$	res 1, 4,	p
11	$\{\neg s, \neg t\}$	res 1, 5,	p



Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$	10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$	11	$\{\neg s, \neg t\}$	res 1, 5, p
3	$\{\neg p, \neg q, \neg r\}$			
4	$\{\neg p, s, t\}$			
5	$\{\neg p, \neg s, \neg t\}$			
6	$\{\neg s, q\}$			
7	$\{r, t\}$			
8	$\{\neg t, s\}$			
9	$\{\neg q, \neg r\}$			res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$	10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$	11	$\{\neg s, \neg t\}$	res 1, 5, p
3	$\{\neg p, \neg q, \neg r\}$	12	$\{\neg s, \neg r\}$	res 6, 9, q
4	$\{\neg p, s, t\}$			
5	$\{\neg p, \neg s, \neg t\}$			
6	$\{\neg s, q\}$			
7	$\{r, t\}$			
8	$\{\neg t, s\}$			
9	$\{\neg q, \neg r\}$			res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$	10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$	11	$\{\neg s, \neg t\}$	res 1, 5, p
3	$\{\neg p, \neg q, \neg r\}$	12	$\{\neg s, \neg r\}$	res 6, 9, q
4	$\{\neg p, s, t\}$			
5	$\{\neg p, \neg s, \neg t\}$			
6	$\{\neg s, q\}$			
7	$\{r, t\}$			
8	$\{\neg t, s\}$			
9	$\{\neg q, \neg r\}$			res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$		
2	$\{\neg p, q, r\}$		
3	$\{\neg p, \neg q, \neg r\}$		
4	$\{\neg p, s, t\}$		
5	$\{\neg p, \neg s, \neg t\}$		
6	$\{\neg s, q\}$		
7	$\{r, t\}$		
8	$\{\neg t, s\}$		
9	$\{\neg q, \neg r\}$	res 1, 3, p	
10	$\{s, t\}$	res 1, 4, p	
11	$\{\neg s, \neg t\}$	res 1, 5, p	
12	$\{\neg s, \neg r\}$	res 6, 9, q	
13	$\{s\}$	res 8, 10, t	

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$	10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$	11	$\{\neg s, \neg t\}$	res 1, 5, p
3	$\{\neg p, \neg q, \neg r\}$	12	$\{\neg s, \neg r\}$	res 6, 9, q
4	$\{\neg p, s, t\}$	13	$\{s\}$	res 8, 10, t
5	$\{\neg p, \neg s, \neg t\}$			
6	$\{\neg s, q\}$			
7	$\{r, t\}$			
8	$\{\neg t, s\}$			
9	$\{\neg q, \neg r\}$			res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$		
2	$\{\neg p, q, r\}$		
3	$\{\neg p, \neg q, \neg r\}$		
4	$\{\neg p, s, t\}$		
5	$\{\neg p, \neg s, \neg t\}$		
6	$\{\neg s, q\}$		
7	$\{r, t\}$		
8	$\{\neg t, s\}$		
9	$\{\neg q, \neg r\}$	res 1, 3, p	
10	$\{s, t\}$	res 1, 4, p	
11	$\{\neg s, \neg t\}$	res 1, 5, p	
12	$\{\neg s, \neg r\}$	res 6, 9, q	
13	$\{s\}$	res 8, 10, t	
14	$\{\neg t\}$	res 11, 13, s	

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$	10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$	11	$\{\neg s, \neg t\}$	res 1, 5, p
3	$\{\neg p, \neg q, \neg r\}$	12	$\{\neg s, \neg r\}$	res 6, 9, q
4	$\{\neg p, s, t\}$	13	$\{s\}$	res 8, 10, t
5	$\{\neg p, \neg s, \neg t\}$	14	$\{\neg t\}$	res 11, 13, s
6	$\{\neg s, q\}$			
7	$\{r, t\}$			
8	$\{\neg t, s\}$			
9	$\{\neg q, \neg r\}$			res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$		
2	$\{\neg p, q, r\}$		
3	$\{\neg p, \neg q, \neg r\}$		
4	$\{\neg p, s, t\}$		
5	$\{\neg p, \neg s, \neg t\}$		
6	$\{\neg s, q\}$		
7	$\{r, t\}$		
8	$\{\neg t, s\}$		
9	$\{\neg q, \neg r\}$	res 1, 3,	p
10	$\{s, t\}$	res 1, 4,	p
11	$\{\neg s, \neg t\}$	res 1, 5,	p
12	$\{\neg s, \neg r\}$	res 6, 9,	q
13	$\{s\}$	res 8, 10,	t
14	$\{\neg t\}$	res 11, 13,	s
15	$\{r\}$	res 7, 14,	t

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$	10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$	11	$\{\neg s, \neg t\}$	res 1, 5, p
3	$\{\neg p, \neg q, \neg r\}$	12	$\{\neg s, \neg r\}$	res 6, 9, q
4	$\{\neg p, s, t\}$	13	$\{s\}$	res 8, 10, t
5	$\{\neg p, \neg s, \neg t\}$	14	$\{\neg t\}$	res 11, 13, s
6	$\{\neg s, q\}$	15	$\{r\}$	res 7, 14, t
7	$\{r, t\}$			
8	$\{\neg t, s\}$			
9	$\{\neg q, \neg r\}$			res 1, 3, p

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$		
2	$\{\neg p, q, r\}$		
3	$\{\neg p, \neg q, \neg r\}$		
4	$\{\neg p, s, t\}$		
5	$\{\neg p, \neg s, \neg t\}$		
6	$\{\neg s, q\}$		
7	$\{r, t\}$		
8	$\{\neg t, s\}$		
9	$\{\neg q, \neg r\}$	res 1, 3, p	
10	$\{s, t\}$	res 1, 4, p	
11	$\{\neg s, \neg t\}$	res 1, 5, p	
12	$\{\neg s, \neg r\}$	res 6, 9, q	
13	$\{s\}$	res 8, 10, t	
14	$\{\neg t\}$	res 11, 13, s	
15	$\{r\}$	res 7, 14, t	
16	$\{\neg r\}$	res 12, 13, s	

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$		
2	$\{\neg p, q, r\}$		
3	$\{\neg p, \neg q, \neg r\}$		
4	$\{\neg p, s, t\}$		
5	$\{\neg p, \neg s, \neg t\}$		
6	$\{\neg s, q\}$		
7	$\{r, t\}$		
8	$\{\neg t, s\}$		
9	$\{\neg q, \neg r\}$	res 1, 3, p	
10	$\{s, t\}$	res 1, 4, p	
11	$\{\neg s, \neg t\}$	res 1, 5, p	
12	$\{\neg s, \neg r\}$	res 6, 9, q	
13	$\{s\}$	res 8, 10, t	
14	$\{\neg t\}$	res 11, 13, s	
15	$\{r\}$	res 7, 14, t	
16	$\{\neg r\}$	res 12, 13, s	

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$		10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$		11	$\{\neg s, \neg t\}$	res 1, 5, p
3	$\{\neg p, \neg q, \neg r\}$		12	$\{\neg s, \neg r\}$	res 6, 9, q
4	$\{\neg p, s, t\}$		13	$\{s\}$	res 8, 10, t
5	$\{\neg p, \neg s, \neg t\}$		14	$\{\neg t\}$	res 11, 13, s
6	$\{\neg s, q\}$		15	$\{r\}$	res 7, 14, t
7	$\{r, t\}$		16	$\{\neg r\}$	res 12, 13, s
8	$\{\neg t, s\}$		17	$\perp$	res 15, 16, r
9	$\{\neg q, \neg r\}$	res 1, 3, p			

Recap: Resolution (2/3)

$$p \wedge (p \rightarrow ((q \vee r) \wedge \neg(q \wedge r))) \wedge (p \rightarrow ((s \vee t) \wedge \neg(s \wedge t))) \wedge \\ \wedge (s \rightarrow q) \wedge (\neg r \rightarrow t) \wedge (t \rightarrow s)$$

1	$\{p\}$		10	$\{s, t\}$	res 1, 4, p
2	$\{\neg p, q, r\}$		11	$\{\neg s, \neg t\}$	res 1, 5, p
3	$\{\neg p, \neg q, \neg r\}$		12	$\{\neg s, \neg r\}$	res 6, 9, q
4	$\{\neg p, s, t\}$		13	$\{s\}$	res 8, 10, t
5	$\{\neg p, \neg s, \neg t\}$		14	$\{\neg t\}$	res 11, 13, s
6	$\{\neg s, q\}$		15	$\{r\}$	res 7, 14, t
7	$\{r, t\}$		16	$\{\neg r\}$	res 12, 13, s
8	$\{\neg t, s\}$		17	$\perp$	res 15, 16, r
9	$\{\neg q, \neg r\}$	res 1, 3, p			unsatisfiable



Recap: Resolution (3/3)

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee s) \wedge (p \vee \neg r) \wedge (r \vee \neg s)$$

- 1 $\{p, q\}$
- 2 $\{\neg p, \neg r\}$
- 3 $\{\neg q, s\}$
- 4 $\{p, \neg r\}$
- 5 $\{r, \neg s\}$



Recap: Resolution (3/3)

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee s) \wedge (p \vee \neg r) \wedge (r \vee \neg s)$$

- 1 $\{p, q\}$
- 2 $\{\neg p, \neg r\}$
- 3 $\{\neg q, s\}$
- 4 $\{p, \neg r\}$
- 5 $\{r, \neg s\}$
- 6 $\{q, \neg r\}$ 1, 2, p



Recap: Resolution (3/3)

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee s) \wedge (p \vee \neg r) \wedge (r \vee \neg s)$$

- 1 $\{p, q\}$
- 2 $\{\neg p, \neg r\}$
- 3 $\{\neg q, s\}$
- 4 $\{p, \neg r\}$
- 5 $\{r, \neg s\}$
- 6 $\{q, \neg r\}$ 1, 2, p
- 7 $\{p, s\}$ 1, 3, q



Recap: Resolution (3/3)

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee s) \wedge (p \vee \neg r) \wedge (r \vee \neg s)$$

- 1 $\{p, q\}$
- 2 $\{\neg p, \neg r\}$
- 3 $\{\neg q, s\}$
- 4 $\{p, \neg r\}$
- 5 $\{r, \neg s\}$
- 6 $\{q, \neg r\}$ 1, 2, p
- 7 $\{p, s\}$ 1, 3, q
- 8 $\{\neg r\}$ 2, 4, p



Recap: Resolution (3/3)

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee s) \wedge (p \vee \neg r) \wedge (r \vee \neg s)$$

- 1 $\{p, q\}$
- 2 $\{\neg p, \neg r\}$
- 3 $\{\neg q, s\}$
- 4 $\{p, \neg r\}$
- 5 $\{r, \neg s\}$
- 6 $\{q, \neg r\}$ 1, 2, p
- 7 $\{p, s\}$ 1, 3, q
- 8 $\{\neg r\}$ 2, 4, p
- 9 $\{\neg p, \neg s\}$ 2, 5, r



Recap: Resolution (3/3)

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee s) \wedge (p \vee \neg r) \wedge (r \vee \neg s)$$

- 1 $\{p, q\}$
- 2 $\{\neg p, \neg r\}$
- 3 $\{\neg q, s\}$
- 4 $\{p, \neg r\}$
- 5 $\{r, \neg s\}$
- 6 $\{q, \neg r\}$ 1, 2, p
- 7 $\{p, s\}$ 1, 3, q
- 8 $\{\neg r\}$ 2, 4, p
- 9 $\{\neg p, \neg s\}$ 2, 5, r
- 10 $\{\neg r, s\}$ 2, 7, p



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- 4 $\{p, \neg r\}$
- 5 $\{r, \neg s\}$
- 6 $\{q, \neg r\}$ 1, 2, p
- 7 $\{p, s\}$ 1, 3, q
- 8 $\{\neg r\}$ 2, 4, p
- 9 $\{\neg p, \neg s\}$ 2, 5, r
- 10 $\{\neg r, s\}$ 2, 7, p
- 11 $\{\neg q, r\}$ 3, 5, s

Recap: Resolution (3/3)

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee s) \wedge (p \vee \neg r) \wedge (r \vee \neg s)$$

1	$\{p, q\}$		12	$\{\neg p, \neg q\}$	3, 9, s	23	$\{q, \neg q\}$	6, 11, r
2	$\{\neg p, \neg r\}$		13	$\{p, \neg s\}$	4, 5, r	24	$\{p, \neg p\}$	7, 9, s
3	$\{\neg q, s\}$		14	$\{\neg r, \neg s\}$	4, 9, p	25	$\{p\}$	7, 19, s
4	$\{p, \neg r\}$		15	$\{p, \neg q\}$	4, 11, r	26	$\{\neg q\}$	8, 11, r
5	$\{r, \neg s\}$		16	$\{\neg q, \neg r\}$	4, 12, p			
6	$\{q, \neg r\}$	1, 2, p	17	$\{q, \neg s\}$	5, 6, r			
7	$\{p, s\}$	1, 3, q	18	$\{p, r\}$	5, 7, s			
8	$\{\neg r\}$	2, 4, p	19	$\{\neg s\}$	5, 8, r			
9	$\{\neg p, \neg s\}$	2, 5, r	20	$\{s, \neg s\}$	5, 10, r			
10	$\{\neg r, s\}$	2, 7, p	21	$\{r, \neg r\}$	5, 10, s			
11	$\{\neg q, r\}$	3, 5, s	22	$\{\neg q, \neg s\}$	5, 16, r			

Recap: Resolution (3/3)

$$(p \vee q) \wedge (\neg p \vee \neg r) \wedge (\neg q \vee s) \wedge (p \vee \neg r) \wedge (r \vee \neg s)$$

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3	$\{\neg q, s\}$		14	$\{\neg r, \neg s\}$	4, 9, p	25	$\{p\}$	7, 19, s
4	$\{p, \neg r\}$		15	$\{p, \neg q\}$	4, 11, r	26	$\{\neg q\}$	8, 11, r
5	$\{r, \neg s\}$		16	$\{\neg q, \neg r\}$	4, 12, p			
6	$\{q, \neg r\}$	1, 2, p	17	$\{q, \neg s\}$	5, 6, r			no further steps
7	$\{p, s\}$	1, 3, q	18	$\{p, r\}$	5, 7, s			
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1 $\{p, q\}$	12 $\{\neg p, \neg q\}$ 3, 9, s	23 $\{q, \neg q\}$ 6, 11, r
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4 $\{p, \neg r\}$	15 $\{p, \neg q\}$ 4, 11, r	26 $\{\neg q\}$ 8, 11, r
5 $\{r, \neg s\}$	16 $\{\neg q, \neg r\}$ 4, 12, p	
6 $\{q, \neg r\}$ 1, 2, p	17 $\{q, \neg s\}$ 5, 6, r	no further steps
7 $\{p, s\}$ 1, 3, q	18 $\{p, r\}$ 5, 7, s	
8 $\{\neg r\}$ 2, 4, p	19 $\{\neg s\}$ 5, 8, r	satisfiable
9 $\{\neg p, \neg s\}$ 2, 5, r	20 $\{s, \neg s\}$ 5, 10, r	
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6 $\{q, \neg r\}$ 1, 2, p	17 $\{q, \neg s\}$ 5, 6, r	no further steps
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Propositional Logic is Not Very Expressive

statements like:

- Every programmer loves every debugging tool
- Some debugging tool is loved by every programmer
- A programmer loves themselves
- No student attended every lecture
- No lecture was attended by every student
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cannot be expressed **adequately** in propositional logic



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cannot be expressed **adequately** in propositional logic

In particular problem with:

- infinite collections (e.g. program inputs)
- transitive verbs (e.g. “loves”)
- relative pronouns (e.g. “whose”)



Some Translations

No emus fly:

$$\forall x(Emu(x) \rightarrow \neg Flies(x))$$



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There are black swans:

$$\exists x(Black(x) \wedge Swan(x))$$



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If all push the cart, the donkey will be happy:

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If the cart is pushed, the donkey will be happy:

$$(\exists x P(x, c)) \rightarrow H(d)$$



Building Blocks

Five new kinds of symbols:

1. constants (e.g. c for “**the** cart”, d for “**the** donkey”)
 - Usually beginning of alphabet, so $a, b, c, d, \dots$



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4. variables (e.g. x, y, z ; end of alphabet)
5. functions (e.g. $f, g, +$)



More Translations

Tina found Rover and returned him to Anne:

Found(tina, rover) $\wedge$ Gave(tina, rover, anne)



More Translations

Tina found Rover and returned him to Anne:

$$\textit{Found}(\textit{tina}, \textit{rover}) \wedge \textit{Gave}(\textit{tina}, \textit{rover}, \textit{anne})$$

Tina found a dog and gave **it** to Anne:

$$\exists x(\textit{Dog}(x) \wedge \textit{Found}(\textit{tina}, x) \wedge \textit{Gave}(\textit{tina}, x, \textit{anne}))$$



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Mothers' mothers are grandmothers:

$$\forall x \forall y \forall z ((\text{Mother}(x, y) \wedge \text{Mother}(y, z)) \rightarrow \text{Grandmother}(x, z))$$